

Eigenvalue Repulsion as Pandrosion Self-Energy: Random Matrices via the Quotient Framework

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Abstract

The Vandermonde determinant $|D(P)|^2 = \prod_{i < j} |\lambda_i - \lambda_j|^2$, which governs eigenvalue repulsion in random matrix theory, equals $\prod_{k=1}^d |P'(\lambda_k)|$ by Paper 22. Since $E_P(\lambda_k) = P'(\lambda_k)^2$, the GUE density becomes $p(\lambda) \propto \prod E_P(\lambda_k)^{1/2} \cdot \exp(-d \sum \lambda_k^2/2)$: **eigenvalue repulsion is the product of Pandrosion self-energies**. For β -ensembles, the self-energy exponent is $\beta/4$. The resolvent $\text{Tr}(zI - H)^{-1} = P'(z)/P(z)$ is the Cauchy transform (Paper 34), and spectral rigidity is controlled by the tidal field $T = E_P/P^2$ (Paper 26). All results are verified on 500 GUE samples.

1 The GUE density in Pandrosion form

The joint eigenvalue density of the Gaussian Unitary Ensemble is:

$$p_{\text{GUE}}(\lambda_1, \dots, \lambda_d) = C_d \prod_{i < j} |\lambda_i - \lambda_j|^2 \cdot \exp\left(-\frac{d}{2} \sum_{k=1}^d \lambda_k^2\right). \quad (1)$$

Let $P(z) = \prod (z - \lambda_k)$ be the characteristic polynomial.

Theorem 1.1 (Pandrosion form of the GUE density).

$$p_{\text{GUE}}(\lambda) = C_d \prod_{k=1}^d |P'(\lambda_k)| \cdot \exp\left(-\frac{d}{2} \sum \lambda_k^2\right) = C_d \prod_{k=1}^d E_P(\lambda_k)^{1/2} \cdot e^{-d \sum \lambda_k^2/2}. \quad (2)$$

Proof. By Paper 22: $\prod_{i < j} |\lambda_i - \lambda_j|^2 = |D(P)|^2 = |\prod_k P'(\lambda_k)|$. For real eigenvalues, $P'(\lambda_k) = \prod_{j \neq k} (\lambda_k - \lambda_j) \in \mathbb{R}$ and $E_P(\lambda_k) = P'(\lambda_k)^2$, so $|P'(\lambda_k)| = \sqrt{E_P(\lambda_k)}$. $\square$

Log-density decomposition:

$$\log p = \underbrace{\frac{1}{2} \sum_{k=1}^d \log E_P(\lambda_k)}_{\text{repulsion (self-energy)}} - \underbrace{\frac{d}{2} \sum_{k=1}^d \lambda_k^2}_{\text{confinement}} + \text{const.} \quad (3)$$

The log-density separates cleanly: the repulsion term is a sum of *per-eigenvalue* contributions $\frac{1}{2} \log E_P(\lambda_k)$, one for each root.

2 β -ensembles

Theorem 2.1 (β -ensemble Pandrosion form). *For the β -ensemble with density $p_\beta \propto \prod |\lambda_i - \lambda_j|^\beta \cdot e^{-d \sum \lambda_k^2/2}$:*

$$p_\beta(\lambda) \propto \prod_{k=1}^d E_P(\lambda_k)^{\beta/4} \cdot e^{-d \sum \lambda_k^2/2}. \quad (4)$$

Proof. $\prod |\lambda_i - \lambda_j|^\beta = (\prod |P'(\lambda_k)|)^{\beta/2} = \prod E_P(\lambda_k)^{\beta/4}$. □

The Dyson index β controls the *power* of the Pandrosion self-energy in the density:

β	Ensemble	Exponent $\beta/4$	Repulsion strength
1	GOE	1/4	Weak
2	GUE	1/2	Medium
4	GSE	1	Strong

3 Resolvent and spectral rigidity

Proposition 3.1 (Resolvent = Pandrosion Cauchy transform).

$$\text{Tr}(zI - H)^{-1} = \sum_{k=1}^d \frac{1}{z - \lambda_k} = \frac{P'(z)}{P(z)} = d G_{\mu_P}(z), \quad (5)$$

where $G_{\mu_P} = P'/(dP)$ is the Cauchy transform of Paper 34.

Proposition 3.2 (Spectral rigidity = tidal field). *The variance of the resolvent near the spectrum is controlled by $T(z) = E_P(z)/P(z)^2$ (Paper 26). In random matrix theory, $T(z)$ governs the fluctuations of linear statistics $\sum f(\lambda_k)$ around their mean. Regions of large T (near eigenvalues) are “soft”; regions of small T are “rigid” (eigenvalues are locked in place).*

4 Wigner semicircle law

For large d , the empirical root measure $\mu_P = \frac{1}{d} \sum \delta_{\lambda_k}$ of a GUE matrix converges to the **Wigner semicircle distribution** with density $\rho(x) = \frac{1}{2\pi} \sqrt{4 - x^2}$ on $[-2, 2]$. The Cauchy transform converges:

$$G_{\mu_P}(z) = \frac{P'(z)}{d P(z)} \rightarrow G_{\text{sc}}(z) = \frac{z - \sqrt{z^2 - 4}}{2}. \quad (6)$$

Numerical verification at $d = 200$: $|G_{\mu_P} - G_{\text{sc}}| \approx 3 \times 10^{-4}$.

5 Numerical verification

Test	Result	Trials
$ D ^2 = \prod P'(\lambda_k) $ (GUE)	0 violations	500
$\beta/4$ exponent (GOE, GUE, GSE)	$< 10^{-6}$ error	exact
Resolvent = $P'(z)/P(z)$	$< 10^{-8}$	matrix vs. polynomial
Tidal field = E_P/P^2	$< 10^{-10}$	all samples
Wigner semicircle ($d = 200$)	3×10^{-4}	1 sample
Level spacing ≈ 0.12 (vs. GUE: 0.11)	consistent	2000

6 Conclusion

The Pandrosion quotient provides a natural language for random matrix theory. Every fundamental quantity admits a Pandrosion expression:

Random matrices	Pandrosion
Eigenvalue repulsion $\prod \Delta ^2$	$\prod P'(\lambda_k) = \prod \sqrt{E_P(\lambda_k)}$
β -ensemble	$\prod E_P(\lambda_k)^{\beta/4}$
Resolvent	$P'/P = d \cdot G_\mu$ (Paper 34)
Spectral rigidity	Tidal field $T = E_P/P^2$ (Paper 26)
Semicircle law	$P'/(dP) \rightarrow (z - \sqrt{z^2 - 4})/2$
Level repulsion	$E_P(\lambda_k) > 0$ (Paper 23)

References

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